

Mastering Quantum Mechanics — Problem 9.4

Generating function for Hermite polynomials

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Problem 9.4

Generating function for Hermite polynomials. The Hermite polynomials are defined by

$$e^{-s^2+2s\xi} = \sum_{n=0}^{\infty} H_n(\xi) \frac{s^n}{n!}. \quad (1)$$

(1) Find the leading term of $H_n(\xi)$. (2) Find operations on (1) that generate $\sum_n n H_n \frac{s^n}{n!}$, $\sum_n H'_n \frac{s^n}{n!}$, $\sum_n H''_n \frac{s^n}{n!}$; combine them to evaluate $\sum_n (H''_n - 2\xi H'_n + 2n H_n) \frac{s^n}{n!}$ and identify the ODE satisfied by H_n .

Hints & Orientation

- (1) The highest power of ξ at order s^n comes purely from $e^{2s\xi}$ (the factor e^{-s^2} carries no ξ , only lowers the available power of s). Match the s^n coefficients.
- (2) Three operations on the generating function $G \equiv e^{-s^2+2s\xi}$: $s \partial_s$ produces the factor n ; ∂_ξ differentiates the H_n ; apply ∂_ξ twice for H''_n . Each acts simply on the closed-form G .
- Linear independence of $\{s^n/n!\}$ turns the vanishing combined sum into the Hermite differential equation term by term.

Solution

(1) Leading term

Write $G = e^{2s\xi} e^{-s^2} = \sum_{k,m} \frac{(2\xi)^k}{k!} s^k \frac{(-1)^m}{m!} s^{2m}$. The coefficient of s^n is $\sum_{k+2m=n} \frac{(2\xi)^k}{k!} \frac{(-1)^m}{m!}$; the highest power of ξ is $k = n$ ($m = 0$), giving $(2\xi)^n/n!$. Matching with $H_n(\xi)/n!$ from (1),

$$\boxed{H_n(\xi) = 2^n \xi^n + (\text{terms of order } \xi^{n-2} \text{ and lower})}. \quad (1)$$

(Consistent with $H_0 = 1$, $H_1 = 2\xi$, $H_2 = 4\xi^2 - 2$, $H_3 = 8\xi^3 - 12\xi$.)

(2) Operations on the generating function

Let $G \equiv e^{-s^2+2s\xi} = \sum_n H_n(\xi) \frac{s^n}{n!}$.

The factor n — apply $s \partial_s$. $s \partial_s \sum_n H_n \frac{s^n}{n!} = \sum_n n H_n \frac{s^n}{n!}$, while $s \partial_s G = s(2\xi - 2s)G$, so

$$\sum_n n H_n(\xi) \frac{s^n}{n!} = (2\xi s - 2s^2) G. \quad (2)$$

H'_n — **apply** ∂_ξ . Since $\partial_\xi G = 2s G$,

$$\sum_n H'_n(\xi) \frac{s^n}{n!} = 2s G. \quad (3)$$

H''_n — **apply** ∂_ξ^2 . Since $\partial_\xi^2 G = 4s^2 G$,

$$\sum_n H''_n(\xi) \frac{s^n}{n!} = 4s^2 G. \quad (4)$$

The combination and the Hermite equation

Adding,

$$\sum_n (H''_n - 2\xi H'_n + 2n H_n) \frac{s^n}{n!} = 4s^2 G - 2\xi(2s G) + 2(2\xi s - 2s^2) G \quad (5)$$

$$= (4s^2 - 4\xi s + 4\xi s - 4s^2) G = 0. \quad (6)$$

Because the powers $s^n/n!$ are linearly independent, each coefficient must vanish:

$$\boxed{H''_n(\xi) - 2\xi H'_n(\xi) + 2n H_n(\xi) = 0}. \quad (7)$$

This is the **Hermite differential equation** (cf. Eq. (9.2.6) with $\mathcal{E} = 2n + 1$).

Checks

- $H_2 = 4\xi^2 - 2$: $H''_2 = 8$, $-2\xi H'_2 = -2\xi(8\xi) = -16\xi^2$, $2 \cdot 2 H_2 = 4(4\xi^2 - 2) = 16\xi^2 - 8$; sum $= 8 - 16\xi^2 + 16\xi^2 - 8 = 0$. ✓
- Leading-term recursion: $2^n \xi^n$ matches the known normalization of H_n .